

DRAWING LIST FOR TEST PACKAGE

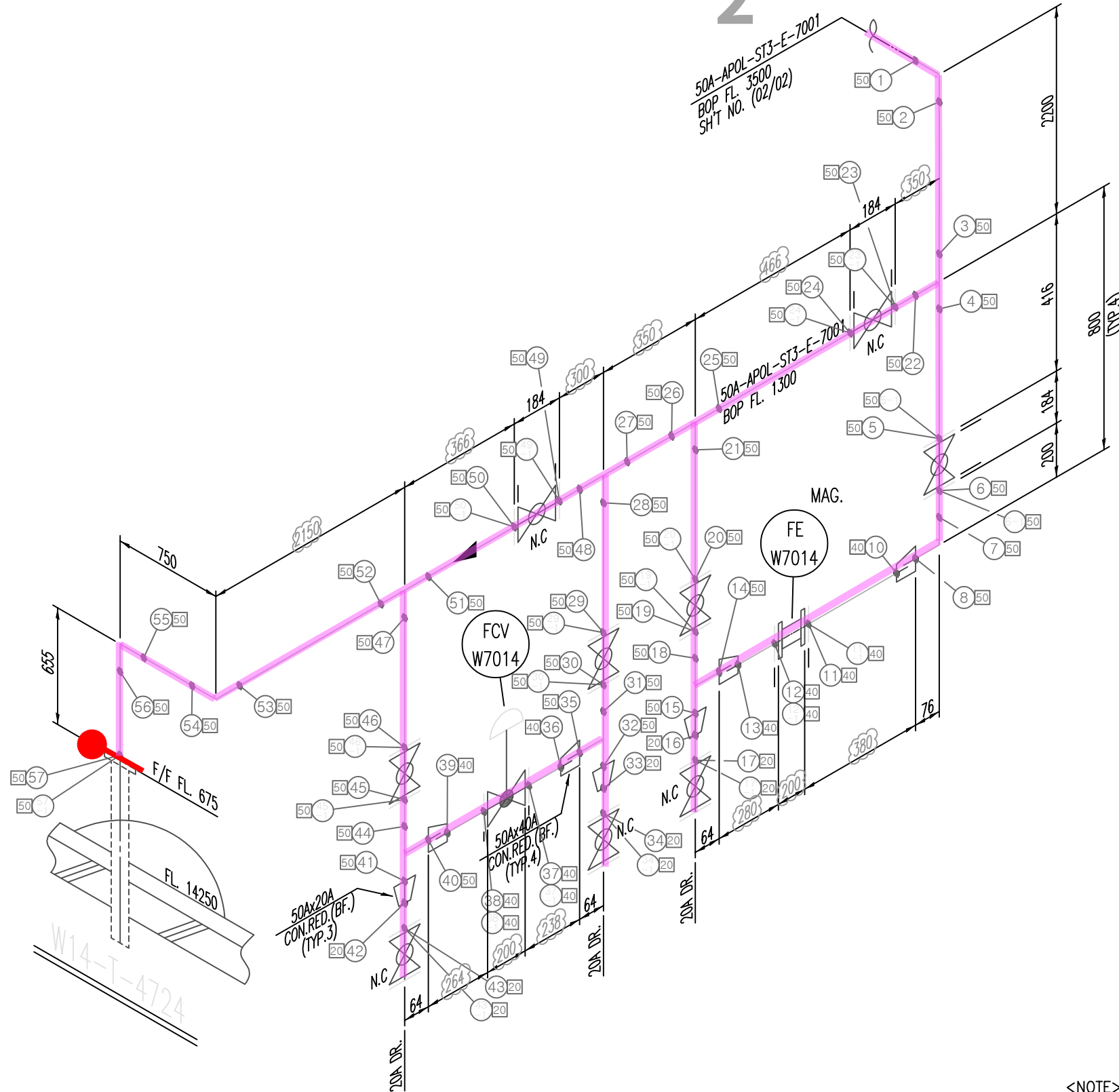
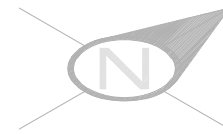
APOL-650

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	650	R2FA	50A-APOL-ST3-E-7001-01	1 OF 2	0	CJ-M15W14-PR-A20-000502		
2	650	R2FA	50A-APOL-ST3-E-7001-02	2 OF 2	0	CJ-M15W14-PR-A20-000149		
3	650	2FA	50A-APOL-ST3-N-7001-01	1 OF 1	0			
4								
5								
6								
7								
8								
9								
10								
11								
12								

2



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA								
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020016-A								
						REF. P. & ID	CJ-M15W14-PR-A20-000502	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	COMMODITY	JOB NO.
														APOL	22330D
														LINE NO.	SH'T NO.
														50A-APOL-ST3-E-7001	01 / 02
															REV.
															0

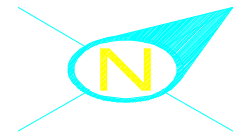
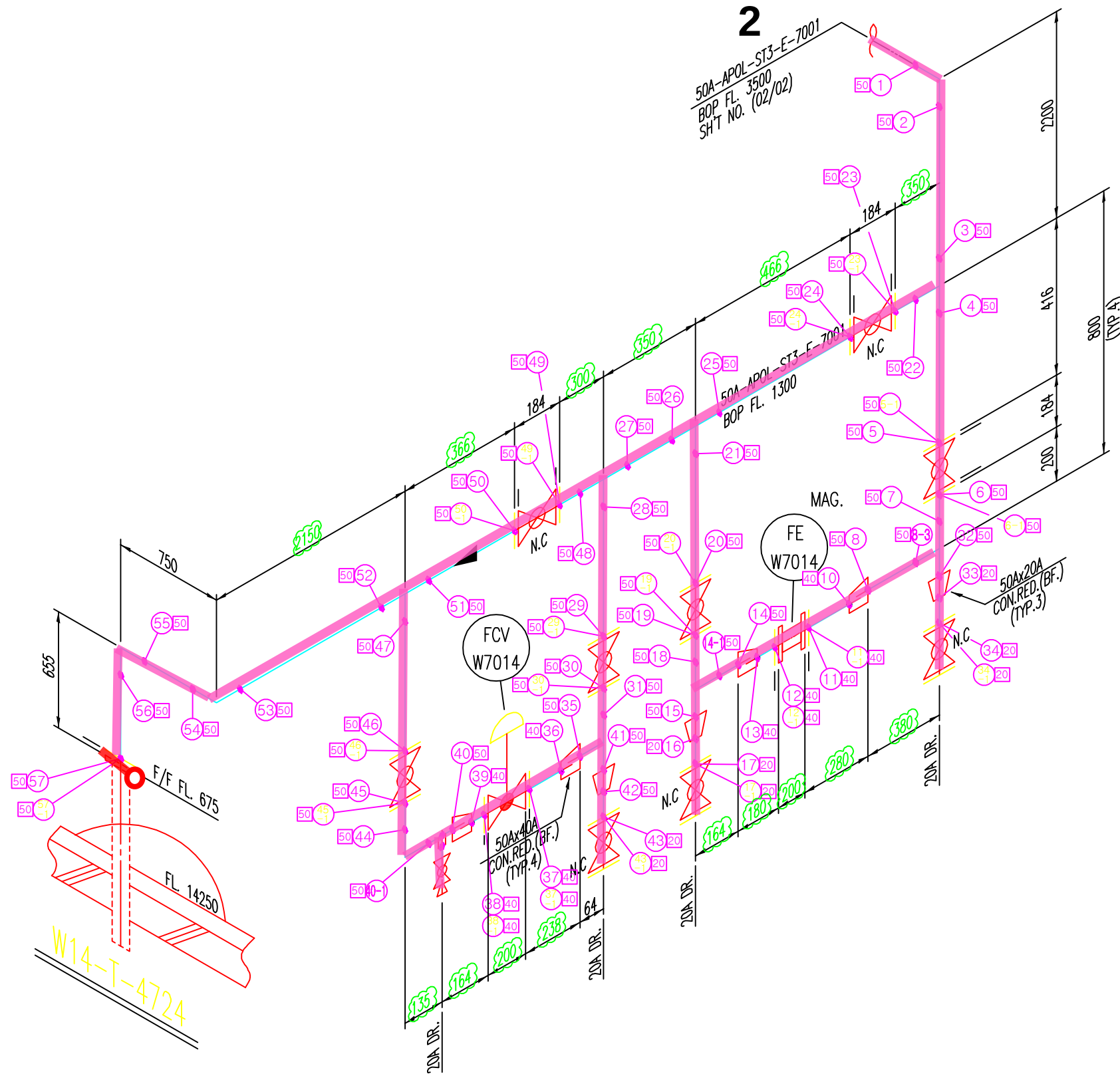


ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	20A	0.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	40A	1.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	50A	10.1M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	4EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	50A	7EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
RC	50Ax20A	3EA	REDUCER CON, STS304, ERW, BE - SCH 10S
RE	50Ax40A	4EA	REDUCER ECC, STS304, ERW, BE - SCH 10S
VB	20A	3EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB	50A	6EA	BALL VALVE, KS 10K, RF FLG'D, BODY-SSC13 / TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	20A	3EA	FLG SO, KS 10K, RF, STS304
F	40A	4EA	FLG SO, KS 10K, RF, STS304
F	50A	13EA	FLG SO, KS 10K, RF, STS304
FB	20A	3EA	FLG BLIND, KS 10K, RF, STS304
G	20A	6EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	13EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	20A	6SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	13SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
LOTO	20A	3EA	LOTO
LOTO	50A	2EA	LOTO
INC	25tX20A	0.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tX40A	1.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tX50A	10.1M	관 보온재(고무발포) + 알루미늄 CASING



1

CLIENT	CONTRACTOR	PROJECT NAME
	SK ecoplant	M15X Ph4 Project
COMMODITY	APOL	JOB NO.
		22330D
LINE NO.	50A-APOL-ST3-E-7001	SH'T NO.
		01 / 02
		REV.
		0



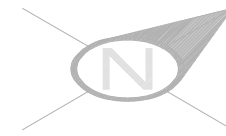
SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	20A	0.4M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	40A	1.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	50A	10.1M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	4EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	50A	7EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
RC	50Ax20A	3EA	REDUCER CON, STS304, ERW, BE - SCH 10S
RE	50Ax40A	4EA	REDUCER ECC, STS304, ERW, BE - SCH 10S
VB	20A	3EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB	50A	6EA	BALL VALVE, KS 10K, RF FLG'D, BODY-SSC13 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB1	20A	2EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
PLG	20A	2EA	PLUG ROUND HEAD, STSF304, SCR'D
F	20A	3EA	FLG SO, KS 10K, RF, STS304
F	40A	4EA	FLG SO, KS 10K, RF, STS304
F	50A	13EA	FLG SO, KS 10K, RF, STS304
FB	20A	3EA	FLG BLIND, KS 10K, RF, STS304
G	20A	6EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	13EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	20A	6SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	13SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
LOTO	20A	3EA	LOTO
LOTO	50A	2EA	LOTO
INC	25tX20A	0.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tX40A	1.3M	관 보온재(고무발포) + 알루미늄 CASING
INC	25tX50A	10.1M	관 보온재(고무발포) + 알루미늄 CASING

<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14												
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA												
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020016-A												
						REF. P. & ID	CJ-M15W14-PR-A20-000502												
						REV. NO.	25.01.24												
						DATE	APPROVED FOR CONSTRUCTION												
						DESCRIPTION	항래중												
						BY	안남후												
						CHK	박동영												
						APP'D													

CLIENT	SK hynix	1
CONTRACTOR	SK ecoplant	PROJECT NAME
		M15X Ph4 Project
COMMODITY	APOL	JOB NO.
		22330D
LINE NO.	50A-APOL-ST3-E-7001	SH'T NO.
		01 / 02
		REV.
		0



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

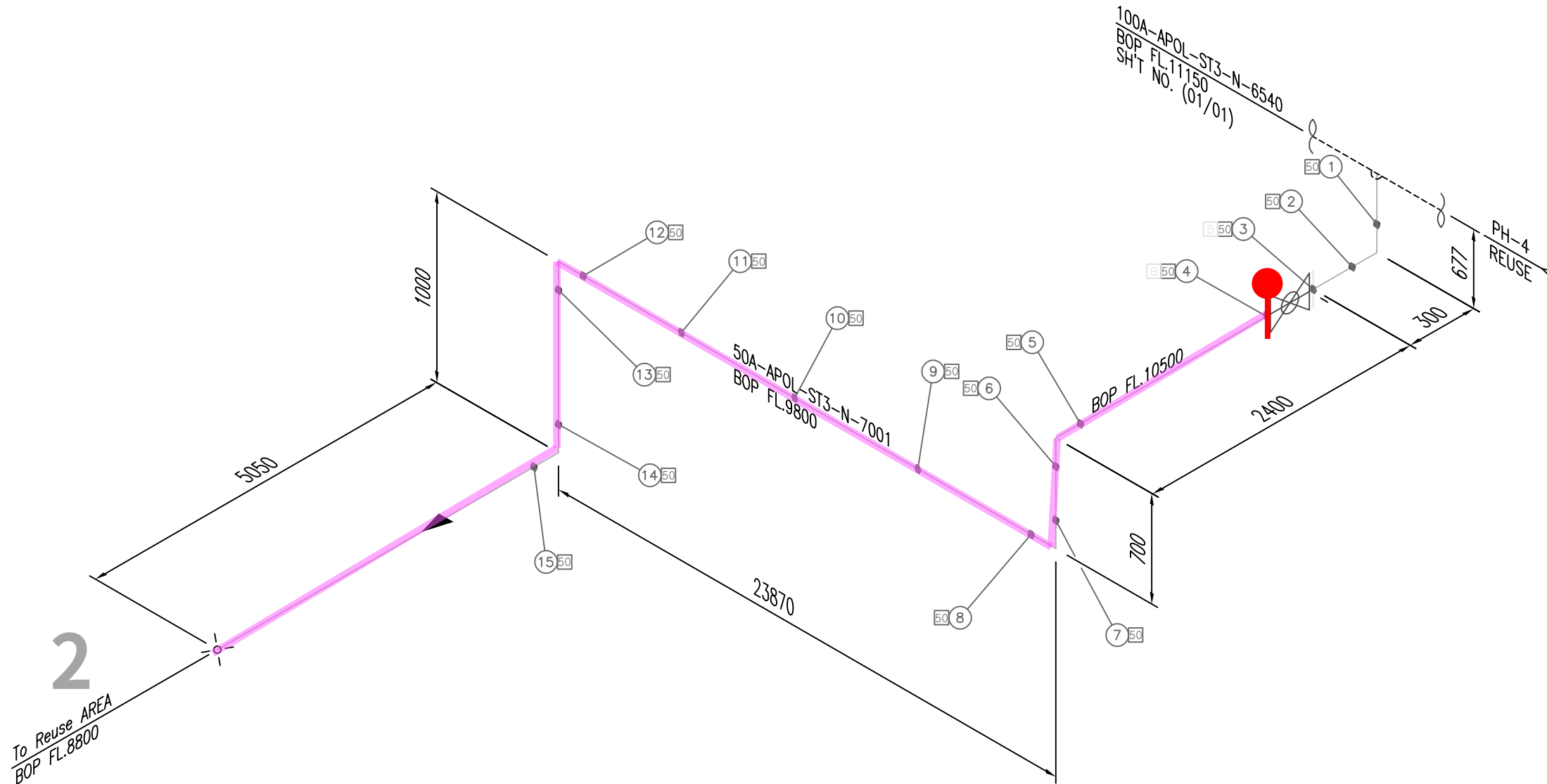
INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA						
			OPERATING	4	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-020016-C							
						REF. P. & ID.			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						CJ-M15W14-PR-A20-000502		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

[illegible]

CLIENT		2	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
APOL		22330D	
LINE NO.		SH'T NO.	REV.
50A-APOL-ST3-E-7001		02 / 02	0



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	50A	34.0M	PIPE, STS 304TP, ERW, BE – SCH 10S
E90	50A	5EA	ELBOW 90 DEG LR, STS304, ERW, BE – SCH 10S
VB	50A	1EA	BALL VALVE, KS 10K, RF, FLG'D, BODY-SSC13 /
			TRIM-ST3304 / SEAT-PTFE, FB, LEVER
F	50A	2EA	FLG SO, KS 10K, RF, STS304
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT		3	
			
CONTRACTOR		PROJECT NAME	
COMMODITY		JOB NO.	
APOL		22330D	
LINE NO.	50A-APOL-ST3-N-7001	SH'T NO.	REV.
		01 / 01	0

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	5.7	60	UNIT	2FA						
			OPERATING	5.1	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-020012-H							
						REF. P. & ID.							
						CJ-M15W14-PR-A20-000149							
									24.12.20	APPROVED FOR CONSTRUCTION	홍재환	안남후	박영호
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP.	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음